

To facilitate the execution of the exercise, a Makefile is used. By running *make info*, the instructions for use are printed.

MATCH/MASK Analysis

To define MATCH, we need to specify the fixed fields of the instruction we want to add. In this case the instruction is of type R and 32-bits, so we need to define the fields opcode[6:0], funct3[14:12] and funct7[31:25]. So we have:

- opcode[1:0] = 0b11 (due to the use of 32-bit and not 16-bit/compressed instructions).
- opcode[6:2] = 0b00010 (since this opcode is not used by another instruction).
- funct3 = 0b000 (it must be unique for each opcode. Since there is no other instruction with the same opcode we can set it as we wish).
- funct7 = 0x00 (for the same reason as funct3).

```
MATCH_MAXX = (funct7 << 25) | (funct3 << 12) | opcode
             = (0x0 << 25)   | (0x0 << 12)   | 0xb
             = 0x0000000b
```

To define MASK, we need to set with 1 the fixed bits of the instruction (those that do not change during runtime). This is done so that the decoder can figure out which instruction is being executed by doing: (insn & MASK_XXXX) == MATCH_XXXX. In this case (32-bit R-type instruction) the fixed fields are opcode[6:0], funct3[14:12] and funct7[31:25]. So we have:

```
MASK_MAXX = (0x7f << 25) | (0x7 << 12) | 0x7f
            = 0xfe00707f
```

Test code

```
#include <stdint.h>
#include <stdio.h>

int main(void) {
    uint32_t rs1 = 3, rs2 = 1, res = 0;

    asm volatile(
        "maxx %0, %1, %2"
        : "=r"(res)
        : "r"(rs1), "r"(rs2)
    );

    printf("Result = %u\n", res);

    return 0;
}
```

The objdump for main

```

spyros@Spyros:~/arch_lab$ make check OUT=./askisi1/test-instructions/maxx.riscv
/opt/riscv/bin/riscv64-unknown-elf-objdump -d ./askisi1/test-
instructions/maxx.riscv | grep -A20 "<main>"
1019e: 04e000ef jal 101ec <main>
101A2: BFBD J 10120 <Exit>

000000000000101a4 <__do_global_dtors_aux>:
101a4:      1141          addi    sp,sp,-16
101a6:      e022          sd      s0,0(sp)
101A8:2501C783 lbu A5,592(GP)#1B5D8 <completed.1>
101ac:      e406          sd      ra,8(sp)
101ae:      ef91          bnez    a5,101ca
<__do_global_dtors_aux+0x26>
101b0:      00000793        li      a5,0
101b4:      cb81          beqz    a5,101c4
<__do_global_dtors_aux+0x20>
101b6:      6569          lui     a0,0x1a
101b8:9a050513 addi a0,a0,-1632#199a0 <__EH_FRAME_BEGIN__>
101bc:      00000097        auipc   ra,0x0
101c0:000000e7 jalr zero # 0 <exit-0x10120>
101c4:      4785          li      a5,1
101C6:24F18823 SB A5,592(GP)#1B5D8 <Completed.1>
101ca:      60a2          ld      ra,8(sp)
101cc:      6402          ld      s0,0(sp)
101ce:      0141          addi    sp,sp,16
101d0:      8082          ret

--
000000000000101ec <main>:
101ec:      1101          addi    sp,sp,-32
101ee:      ec06          sd      ra,24(sp)
101f0:      e822          sd      s0,16(sp)
101f2:      1000          addi    s0,sp,32
101f4:      478d          li      a5,3
101f6:      fef42623        sw      a5,-20(s0)
101fa:      4785          li      a5,1
101fc:      fef42423        sw      a5,-24(s0)
10200:      fe042223        sw      zero,-28(s0)
10204:      fec42783        lw      a5,-20(s0)
10208:      fe842703        lw      a4,-24(s0)
1020c:      00e7878b        maxx   a5,a5,a4
10210:      fef42223        sw      a5,-28(s0)
10214:      fe442783        lw      a5,-28(s0)
10218:      85be          mv      a1,a5

```

```

1021a:    67e5                lui    a5,0x19
1021c:    1a078513             addi   a0,a5,416 # 191a0 <__clzdi2+0x3c>
10220:    328000EF JAL 10548 <PRINTF>
10224:    4781                li     a5,0
10226:    853e                mv     a0,a5

```

The gem5 output

```

spyros@Spyros:~/arch_lab$ make run OUT=./askisi1/test-instructions/maxx.riscv
/home/spyros/gem5/build/RISCV/gem5.opt \
/home/spyros/gem5/configs/deprecated/example/se.py \
--cpu-type=DerivO3CPU \
--caches --l2cache \
--cmd=/home/spyros/arch_lab/./askisi1/test-instructions/maxx.riscv
gem5 Simulator System.  https://www.gem5.org
gem5 is copyrighted software; use the --copyright option for details.

gem5 version 25.1.0.0
gem5 compiled Mar 27 2026 19:07:20
gem5 started Mar 27 2026 20:58:54
gem5 executing on Spyros, pid 526269
instruction line: /home/spyros/gem5/build/RISCV/gem5.opt
/home/spyros/gem5/configs/deprecated/example/se.py --cpu-type=DerivO3CPU --caches
--l2cache --cmd=/home/spyros/arch_lab/./askisi1/test-instructions/maxx.riscv

warn: The se.py script is deprecated. It will be removed in future releases
of gem5.
Global frequency set at 1000000000000 ticks per second
src/mem/dram_interface.cc:690: warn: DRAM device capacity (8192 Mbytes) does not
match the address range assigned (512 Mbytes)
src/arch/riscv/isa.cc:321: info: RVV enabled, VLEN = 256 bits, ELEN = 64 bits
src/arch/riscv/linux/se_workload.cc:73: warn: Unknown operating system; assuming
Linux.
src/base/statistics.hh:279: warn: One of the stats is a legacy stat. Legacy stat
is a stat that does not belong to any statistics::Group. Legacy stat is
deprecated.
src/base/stats/storage.hh:278: warn: Bucket size (4) does not divide range [1:32]
into equal-sized buckets. Rounding up.
src/base/stats/storage.hh:278: warn: Bucket size (4) does not divide range [1:8]
into equal-sized buckets. Rounding up.
system.remote_gdb: Listening for connections on port 7000
**** REAL SIMULATION ****
Result = 3
Exiting @ tick 8698500 because exiting with last active thread context

```

Description of steps

The implementation steps are:

1. Find values for MATCH and MASK.
2. Add define MATCH and MASK into the *riscv-opc.h* file of the riscv-toolchain.
3. Add DECLARE_INSN into the *riscv-opc.h* file of the riscv-toolchain.
4. Add the instruction in the *riscv-opc.c* file of the riscv-toolchain.
5. Rebuild the RISC-V toolchain (make toolchain JOBS=N) so that the new instruction is recognized when compiling programs.
6. Add the instruction into the *decoder.isa* file so that decoding and correct execution can be done (we define the arithmetic operation to be executed) during the simulation.
7. Rebuild gem5 (make gem5 JOBS=N) to use the new decoder during simulation.
8. Compilation of the code (make build SRC=<path_to_code>/maxx.c).
9. Execute the code (make run OUT=<path_to_compiled_file>/maxx.riscv).

Optional Extension

MATCH/MASK Analysis

For the instruction "minx rd, rs1, rs2" the same steps were followed with maxx. It has the same opcode and funct7 and differs only in funct3 which is 0b001 instead of 0b000. The mask is the same since it is a instruction of the same type. So we have:

```
MATCH_MINX 0x100b
MASK_MINX  0xfe00707f
```

Test code

```
#include <stdint.h>
#include <stdio.h>

int main(void) {
    uint32_t rs1 = 3, rs2 = 1, res = 0;

    asm volatile(
        "minx %0, %1, %2"
        : "=r"(res)
        : "r"(rs1), "r"(rs2)
    );

    printf("Result = %u\n", res);

    return 0;
}
```

The objdump for main

```

spyros@Spyros:~/arch_lab$ make check OUT=./askisi1/test-instructions/minx.riscv
/opt/riscv/bin/riscv64-unknown-elf-objdump -d ./askisi1/test-
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 101b8:9a050513 addi a0,a0,-1632#199a0 <__EH_FRAME_BEGIN__>
101bc:      00000097        auipc   ra,0x0
 101c0:000000e7 jalr zero # 0 <exit-0x10120>
101c4:      4785          li      a5,1
 101C6:24F18823 SB A5,592(GP)#1B5D8 <Completed.1>
101ca:      60a2          ld      ra,8(sp)
101cc:      6402          ld      s0,0(sp)
101ce:      0141          addi    sp,sp,16
101d0:      8082          ret

--
000000000000101ec <main>:
101ec:      1101          addi    sp,sp,-32
101ee:      ec06          sd      ra,24(sp)
101f0:      e822          sd      s0,16(sp)
101f2:      1000          addi    s0,sp,32
101f4:      478d          li      a5,3
101f6:      fef42623        sw      a5,-20(s0)
101fa:      4785          li      a5,1
101fc:      fef42423        sw      a5,-24(s0)
10200:      fe042223        sw      zero,-28(s0)
10204:      fec42783        lw      a5,-20(s0)
10208:      fe842703        lw      a4,-24(s0)
1020c:      00e7978b        minx    a5,a5,a4
10210:      fef42223        sw      a5,-28(s0)
10214:      fe442783        lw      a5,-28(s0)
10218:      85be          mv      a1,a5

```

```

1021a:    67e5                lui    a5,0x19
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10220:    328000EF JAL 10548 <PRINTF>
10224:    4781                li     a5,0
10226:    853e                mv     a0,a5

```

The gem5 output

```

spyros@Spyros:~/arch_lab$ make run OUT=./askisi1/test-instructions/minx.riscv
/home/spyros/gem5/build/RISCV/gem5.opt \
/home/spyros/gem5/configs/deprecated/example/se.py \
--cpu-type=DerivO3CPU \
--caches --l2cache \
--cmd=/home/spyros/arch_lab/./askisi1/test-instructions/minx.riscv
gem5 Simulator System.  https://www.gem5.org
gem5 is copyrighted software; use the --copyright option for details.

gem5 version 25.1.0.0
gem5 compiled Mar 27 2026 19:07:20
gem5 started Mar 27 2026 21:14:23
gem5 executing on Spyros, pid 529746
instruction line: /home/spyros/gem5/build/RISCV/gem5.opt
/home/spyros/gem5/configs/deprecated/example/se.py --cpu-type=DerivO3CPU --caches
--l2cache --cmd=/home/spyros/arch_lab/./askisi1/test-instructions/minx.riscv

warn: The se.py script is deprecated. It will be removed in future releases
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Global frequency set at 1000000000000 ticks per second
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src/base/statistics.hh:279: warn: One of the stats is a legacy stat. Legacy stat
is a stat that does not belong to any statistics::Group. Legacy stat is
deprecated.
src/base/stats/storage.hh:278: warn: Bucket size (4) does not divide range [1:32]
into equal-sized buckets. Rounding up.
src/base/stats/storage.hh:278: warn: Bucket size (4) does not divide range [1:8]
into equal-sized buckets. Rounding up.
system.remote_gdb: Listening for connections on port 7000
**** REAL SIMULATION ****
Result = 1
Exiting @ tick 8698500 because exiting with last active thread context

```

[Description of steps](#)

The steps are the same as those of the maxx instruction with the only difference being the operation described in the decoder.isa file:

```
Rd = (Rs1_ud < Rs2_ud) ? Rs1_ud : Rs2_ud;
```